

Biotechnology English Participant Workbook

Practice pages for realistic field-specific meetings, pushback, documentation, and role-play preparation

Audience: biotech scientists, translational researchers, assay-development teams, platform teams, program managers, alliance managers, business-development staff, and biotech executives

Focus: A high-level biotechnology English curriculum for platform science, assay design, translational evidence, biomarkers, IP, partnerships, investor updates, regulatory-adjacent planning, and scientific-business dialogue.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

How to Use This Workbook

For each module, read a workplace scenario and complete a bounded dialogue. Choose one of four options, then use the answer key to see why the selected phrase fits the evidence, risk, owner, and next step.

Module 1. Platform Technology and Scientific Thesis

Situation

A founder wants to describe the platform as broadly validated.

Stakeholder pressure: Use the strongest possible investor language.

Constraint: The evidence supports specific models and indications, not every future use case.

Terms to use

- platform
- modality
- proof of concept
- translatability

Three-step decision practice

Situation

A founder wants to describe the platform as broadly validated.

Speaker	Line
Stakeholder	Use the strongest possible investor language.
ESL learner	Before we act, I need to confirm the _____ and how it affects modality.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a platform evidence statement.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects modality.

Option	Phrase
A	operating constraint
B	decision owner
C	platform
D	approval path

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Use the strongest possible investor language.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I disagree, but I cannot name the boundary or the decision the group needs to make.
B	I understand the urgency. I need to verify platform and modality against the stated constraint before I recommend the next step.
C	We should act now and resolve platform after the result is visible.
D	A platform evidence statement is unnecessary; verbal assurance should be enough.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A founder wants to describe the platform as broadly validated. - what should happen next?

Option	Phrase
A	Review the stated constraint with the accountable owner, then prepare the platform evidence statement with the evidence, tradeoff, and requested decision.
B	Send every available detail to senior leaders without identifying the decision they need to make.
C	Wait for the problem to become visible before assigning an owner or evidence source.
D	evidence threshold

Module 2. Assay Development and Reproducibility

Situation

A partner asks whether a biomarker assay is ready for decision-making.

Stakeholder pressure: Say the assay works because the pilot was positive.

Constraint: Precision, sensitivity, specificity, controls, and reproducibility need review.

Terms to use

- assay
- sensitivity
- specificity
- reproducibility

Three-step decision practice

Situation

A partner asks whether a biomarker assay is ready for decision-making.

Speaker	Line
Stakeholder	Say the assay works because the pilot was positive.
ESL learner	Before we act, I need to confirm the _____ and how it affects sensitivity.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a assay readiness note.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects sensitivity.

Option	Phrase
A	specificity
B	reproducibility
C	assay
D	sensitivity

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Say the assay works because the pilot was positive.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	operating constraint
B	I understand the urgency. I need to verify assay and sensitivity against the stated constraint before I recommend the next step.
C	evidence threshold
D	approval path

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A partner asks whether a biomarker assay is ready for decision-making. - what should happen next?

Option	Phrase
A	Review the stated constraint with the accountable owner, then prepare the assay readiness note with the evidence, tradeoff, and requested decision.
B	operating constraint
C	decision owner
D	Accept the request immediately because the stakeholder has already expressed urgency.

Module 3. Biomarkers and Translational Strategy

Situation

A team wants to use a biomarker as the main development rationale.

Stakeholder pressure: Make the biomarker the centerpiece of the clinical story.

Constraint: Biological plausibility, patient relevance, and validation status are different.

Terms to use

- biomarker
- patient stratification
- surrogate endpoint

- validation

Three-step decision practice

Situation

A team wants to use a biomarker as the main development rationale.

Speaker	Line
Stakeholder	Make the biomarker the centerpiece of the clinical story.
ESL learner	Before we act, I need to confirm the _____ and how it affects patient stratification.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a biomarker caveat memo.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects patient stratification.

Option	Phrase
A	patient stratification
B	surrogate endpoint
C	validation
D	biomarker

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Make the biomarker the centerpiece of the clinical story.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I understand the urgency. I need to verify biomarker and patient stratification against the stated constraint before I recommend the next step.
B	evidence threshold
C	approval path
D	operating constraint

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A team wants to use a biomarker as the main development rationale. - what should happen next?

Option	Phrase
A	Accept the request immediately because the stakeholder has already expressed urgency.
B	Send every available detail to senior leaders without identifying the decision they need to make.

Option	Phrase
C	Review the stated constraint with the accountable owner, then prepare the biomarker caveat memo with the evidence, tradeoff, and requested decision.
D	decision owner

Module 4. Preclinical Data Packages

Situation

A board member asks whether animal data prove human efficacy.

Stakeholder pressure: Say the model predicts human response.

Constraint: Model relevance, dose, exposure, safety margin, and limitations must be named.

Terms to use

- in vitro
- in vivo
- toxicity
- safety margin

Three-step decision practice

Situation

A board member asks whether animal data prove human efficacy.

Speaker	Line
Stakeholder	Say the model predicts human response.
ESL learner	Before we act, I need to confirm the _____ and how it affects in vivo.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a preclinical evidence summary.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects in vivo.

Option	Phrase
A	in vivo
B	toxicity
C	in vitro
D	decision owner

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Say the model predicts human response.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I understand the urgency. I need to verify in vitro and in vivo against the stated constraint before I recommend the next step.
B	I disagree, but I cannot name the boundary or the decision the group needs to make.
C	evidence threshold
D	approval path

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A board member asks whether animal data prove human efficacy. - what should happen next?

Option	Phrase
A	decision owner
B	Accept the request immediately because the stakeholder has already expressed urgency.
C	Review the stated constraint with the accountable owner, then prepare the preclinical evidence summary with the evidence, tradeoff, and requested decision.
D	operating constraint

Module 5. CMC and Scale-Up in Biotech

Situation

A program lead wants to promise rapid scale-up after financing.

Stakeholder pressure: Assure investors that scale-up will be simple.

Constraint: Yield, process control, comparability, and supply-chain risk are unresolved.

Terms to use

- CMC
- yield
- comparability
- tech transfer

Three-step decision practice

Situation

A program lead wants to promise rapid scale-up after financing.

Speaker	Line
Stakeholder	Assure investors that scale-up will be simple.
ESL learner	Before we act, I need to confirm the _____ and how it affects yield.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a scale-up risk update.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects yield.

Option	Phrase
A	CMC
B	approval path
C	operating constraint
D	decision owner

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Assure investors that scale-up will be simple.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	operating constraint
B	decision owner
C	I understand the urgency. I need to verify CMC and yield against the stated constraint before I recommend the next step.
D	approval path

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A program lead wants to promise rapid scale-up after financing. - what should happen next?

Option	Phrase
A	evidence threshold
B	approval path
C	Review the stated constraint with the accountable owner, then prepare the scale-up risk update with the evidence, tradeoff, and requested decision.
D	Wait for the problem to become visible before assigning an owner or evidence source.

Module 6. IP, Freedom to Operate, and Collaboration

Situation

A collaborator requests broad access to unpublished methods.

Stakeholder pressure: Share the materials to keep the relationship warm.

Constraint: Confidentiality, background IP, publication rights, and freedom to operate matter.

Terms to use

- IP
- freedom to operate
- material transfer agreement

- publication rights

Three-step decision practice

Situation

A collaborator requests broad access to unpublished methods.

Speaker	Line
Stakeholder	Share the materials to keep the relationship warm.
ESL learner	Before we act, I need to confirm the _____ and how it affects freedom to operate.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a collaboration boundary response.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects freedom to operate.

Option	Phrase
A	evidence threshold
B	IP
C	material transfer agreement
D	publication rights

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Share the materials to keep the relationship warm.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I understand the urgency. I need to verify IP and freedom to operate against the stated constraint before I recommend the next step.
B	I disagree, but I cannot name the boundary or the decision the group needs to make.
C	evidence threshold
D	approval path

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A collaborator requests broad access to unpublished methods. - what should happen next?

Option	Phrase
A	Accept the request immediately because the stakeholder has already expressed urgency.
B	Review the stated constraint with the accountable owner, then prepare the collaboration boundary response with the evidence, tradeoff, and requested decision.

Option	Phrase
C	operating constraint
D	decision owner

Module 7. Investor and Board Updates

Situation

The board wants a clean milestone story before a financing round.

Stakeholder pressure: Remove uncertainties from the update.

Constraint: Credibility requires clear milestones, risks, assumptions, and mitigation.

Terms to use

- milestone
- runway
- inflection point
- risk mitigation

Three-step decision practice

Situation

The board wants a clean milestone story before a financing round.

Speaker	Line
Stakeholder	Remove uncertainties from the update.
ESL learner	Before we act, I need to confirm the _____ and how it affects runway.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a investor-ready milestone update.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects runway.

Option	Phrase
A	approval path
B	milestone
C	risk mitigation
D	evidence threshold

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Remove uncertainties from the update.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	evidence threshold
B	I understand the urgency. I need to verify milestone and runway against the stated constraint before I recommend the next step.
C	A investor-ready milestone update is unnecessary; verbal assurance should be enough.
D	I disagree, but I cannot name the boundary or the decision the group needs to make.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - The board wants a clean milestone story before a financing round. - what should happen next?

Option	Phrase
A	Wait for the problem to become visible before assigning an owner or evidence source.
B	evidence threshold
C	Review the stated constraint with the accountable owner, then prepare the investor-ready milestone update with the evidence, tradeoff, and requested decision.
D	Send every available detail to senior leaders without identifying the decision they need to make.

Module 8. Partnering and Business Development

Situation

A potential partner wants exclusive rights after limited data review.

Stakeholder pressure: Agree quickly before they lose interest.

Constraint: Diligence, territory, field, economics, governance, and data rights need definition.

Terms to use

- term sheet
- option
- exclusivity
- due diligence

Three-step decision practice

Situation

A potential partner wants exclusive rights after limited data review.

Speaker	Line
Stakeholder	Agree quickly before they lose interest.
ESL learner	Before we act, I need to confirm the _____ and how it affects option.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a partnering negotiation brief.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects option.

Option	Phrase
A	evidence threshold
B	approval path
C	operating constraint
D	term sheet

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Agree quickly before they lose interest.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I understand the urgency. I need to verify term sheet and option against the stated constraint before I recommend the next step.
B	decision owner
C	We should act now and resolve term sheet after the result is visible.
D	A partnering negotiation brief is unnecessary; verbal assurance should be enough.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A potential partner wants exclusive rights after limited data review. - what should happen next?

Option	Phrase
A	Send every available detail to senior leaders without identifying the decision they need to make.
B	Review the stated constraint with the accountable owner, then prepare the partnering negotiation brief with the evidence, tradeoff, and requested decision.
C	decision owner
D	Accept the request immediately because the stakeholder has already expressed urgency.

Answer Key and Rationale

Check each stage after completing the practice sequence. The rationale explains the decision skill the strongest response demonstrates.

Platform Technology and Scientific Thesis - 1. Complete the field language

Correct answer	Why it fits
C. platform	'platform' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Platform Technology and Scientific Thesis - 2. Choose the strongest professional response

Correct answer	Why it fits
B. I understand the urgency. I need to verify platform and modality against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Platform Technology and Scientific Thesis - 3. Select the next decision move

Correct answer	Why it fits
A. Review the stated constraint with the accountable owner, then prepare the platform evidence statement with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Assay Development and Reproducibility - 1. Complete the field language

Correct answer	Why it fits
C. assay	'assay' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Assay Development and Reproducibility - 2. Choose the strongest professional response

Correct answer	Why it fits
B. I understand the urgency. I need to verify assay and sensitivity against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Assay Development and Reproducibility - 3. Select the next decision move

Correct answer	Why it fits
A. Review the stated constraint with the accountable owner, then prepare the assay readiness note with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Biomarkers and Translational Strategy - 1. Complete the field language

Correct answer	Why it fits
D. biomarker	'biomarker' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Biomarkers and Translational Strategy - 2. Choose the strongest professional response

Correct answer	Why it fits
A. I understand the urgency. I need to verify biomarker and patient stratification against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Biomarkers and Translational Strategy - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the biomarker caveat memo with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Preclinical Data Packages - 1. Complete the field language

Correct answer	Why it fits
C. in vitro	'in vitro' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Preclinical Data Packages - 2. Choose the strongest professional response

Correct answer	Why it fits
A. I understand the urgency. I need to verify in vitro and in vivo against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Preclinical Data Packages - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the preclinical evidence summary with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

CMC and Scale-Up in Biotech - 1. Complete the field language

Correct answer	Why it fits
A. CMC	'CMC' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

CMC and Scale-Up in Biotech - 2. Choose the strongest professional response

Correct answer	Why it fits
C. I understand the urgency. I need to verify CMC and yield against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

CMC and Scale-Up in Biotech - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the scale-up risk update with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

IP, Freedom to Operate, and Collaboration - 1. Complete the field language

Correct answer	Why it fits
B. IP	'IP' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

IP, Freedom to Operate, and Collaboration - 2. Choose the strongest professional response

Correct answer	Why it fits
A. I understand the urgency. I need to verify IP and freedom to operate against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

IP, Freedom to Operate, and Collaboration - 3. Select the next decision move

Correct answer	Why it fits
B. Review the stated constraint with the accountable owner, then prepare the collaboration boundary response with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Investor and Board Updates - 1. Complete the field language

Correct answer	Why it fits
B. milestone	'milestone' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Investor and Board Updates - 2. Choose the strongest professional response

Correct answer	Why it fits
B. I understand the urgency. I need to verify milestone and runway against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Investor and Board Updates - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the investor-ready milestone update with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Partnering and Business Development - 1. Complete the field language

Correct answer	Why it fits
D. term sheet	'term sheet' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Partnering and Business Development - 2. Choose the strongest professional response

Correct answer	Why it fits
A. I understand the urgency. I need to verify term sheet and option against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Partnering and Business Development - 3. Select the next decision move

Correct answer	Why it fits
B. Review the stated constraint with the accountable owner, then prepare the partnering negotiation brief with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.